## Data sources

**JEPX spot prices**
day-ahead, 9 areas, 30-min slots

**JMA-MSM runs**
archived cycles pinned by dissemination time

**ERA5 reanalysis**
day-lagged fields and realised weather

**OCCTO actuals**
recorded PV generation and curtailment

One expanding-window daily walk-forward · 732 daily walk-forward test days × 9 areas · 315,414 half-hour slots
XGBoost carries the ladder; LEAR and naive references · identical folds and one price-only baseline for every tier

### A. Per-area point accuracy
ΔMAE vs price-only

**+0.047 JPY/kWh**
from the stale tier; pooled over 732 days, DM-significant in 7 of 9 areas (Holm-adjusted)

### B. Vintage ladder and gate timing
five tiers, one sample

**+0.206 pre-gate**
0.047→0.253 JPY/kWh; excluded run equivalent (pooled MAE, 1 tick, p = 0.024); +0.200 over the enriched baseline

### C. Two targets, physical vs price
696 common days

**+59.9 MW post-gate**
scaled MAE vs PV potential: 878→819 MW [49.6, 70.6]; residual variance: 14.0% physical, 1.2% price

### D. Curtailment channel
floor slots vs OCCTO

**19–96× more likely**
near-floor prices when PV curtailment is active; 8 areas, 279,590 slots; co-occurrence, not causation